

Forest Cover Type Classification Report

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Machine Learning Project Report

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1 Executive Summary

This report presents an end-to-end machine learning pipeline for the Forest Cover Type classification task, where the objective is to predict one of seven forest cover classes from cartographic variables. The labelled dataset contains 15,120 observations and is perfectly balanced across classes, which makes accuracy an appropriate primary evaluation metric. Exploratory analysis showed that the data are already high quality, with no missing values or duplicate rows, so the work focused on representation, feature engineering, and model selection rather than heavy cleaning.

The strongest gains came from domain-aware feature engineering and the choice of model family. We created a richer representation of the terrain by adding hydrology-based distance features, hillshade aggregates, nonlinear elevation and slope terms, and higher-level ecological descriptors derived from soil and wilderness indicators. An ablation study showed that nonlinear terrain features and hydrology features contributed the most, improving performance meaningfully over the cleaned baseline.

Model comparison confirmed that tree-based ensembles are much better suited to this dataset than linear models. Among the tested approaches, LightGBM, XGBoost, and ExtraTrees performed best and were tuned with Optuna before being combined in a soft voting ensemble. This final ensemble achieved about 0.90 accuracy on the labelled holdout set and 0.86 on the Kaggle public leaderboard, suggesting strong generalisation with limited overfitting. Overall, the project delivers a robust and interpretable predictive pipeline, while also showing that careful feature design and ensemble learning matter more here than complex preprocessing or deep learning by default.

2 Problem Framing and Evaluation Metric

The task is a supervised multiclass classification problem, where the goal is to predict the forest cover type (`Cover_Type`) among seven possible classes. Model development and comparison are primarily based on classification accuracy, which aligns with the competition's ranking metric.

Accuracy is an appropriate primary metric in this setting because the labelled training dataset is perfectly balanced across all seven classes. As a result, accuracy is not artificially inflated by majority-class dominance, and each misclassification contributes approximately equally to the overall score. This makes accuracy a reliable summary measure during model selection.

However, accuracy alone is not sufficient for diagnostic analysis. To better understand model behaviour, we also report a full classification report and confusion matrix on the outer holdout split. These additional metrics allow us to examine class-specific precision and recall and to identify systematic confusions between similar forest types, even when aggregate accuracy appears strong.

3 Exploratory Data Analysis

3.1 Dataset Overview

The labelled training set contains 15,120 observations and 56 columns. The predictors combine continuous cartographic variables with two sparse one-hot indicator blocks, `Wilderness_Area1-4` and `Soil_Type1-40`. The target variable `Cover_Type` has seven classes.

Table 1: Dataset summary.

Item	Value
Number of observations	15,120
Number of variables	56 total (including target and Id)
Feature families	10 continuous terrain/light variables, 4 wilderness indicators, 40 soil indicators
Variable types	12 numeric, 44 categorical
Missing data	0 cells (0.0%)
Duplicate rows	0 rows (0.0%)
Target	`Cover_Type` with 7 classes
Target balance	Perfectly balanced: 2,160 rows per class (14.3% each)

3.2 Data Quality Assessment

The profiling report indicates a clean labelled training set with no missing values and no duplicate rows. This allows us to avoid imputation, row filtering, and missingness indicator engineering. That simplification is important because it lets us focus engineering effort on signal creation instead of repair.

One structural validation step is particularly important, as several engineered features rely on decoding the one-hot encoded blocks. We verified one-of-K for the wilderness variables and soil indicators. In both cases, the row-wise sum equals 1 for every observation (unique row sums: $\{1\}$), confirming that each sample belongs to exactly one category.

3.3 Target Distribution

The target is perfectly balanced in the labelled training set. This is visible in the profiling report and remains a foundational assumption for our metric choice and split design.

This balance has three direct implications for our pipeline design:

- Accuracy is a valid optimisation metric during tuning on labelled splits.
- Class weighting is useful as a robustness choice during tuning, but it is not strictly required to correct label imbalance in the labelled train set.
- Macro and weighted averages are close in the labelled validation setting, so per-class metrics become easier to interpret.

3.4 Sparse Indicator Structure and Ecological Proxies

The wilderness and soil variables are one-hot encoded indicator blocks. This structure is informative but sparse, particularly for certain soil types. While these indicators can serve as strong ecological proxies for cover type, some columns are rarely active, meaning only a limited number of samples inform their effect.

Wilderness area prevalence in the labelled training set:

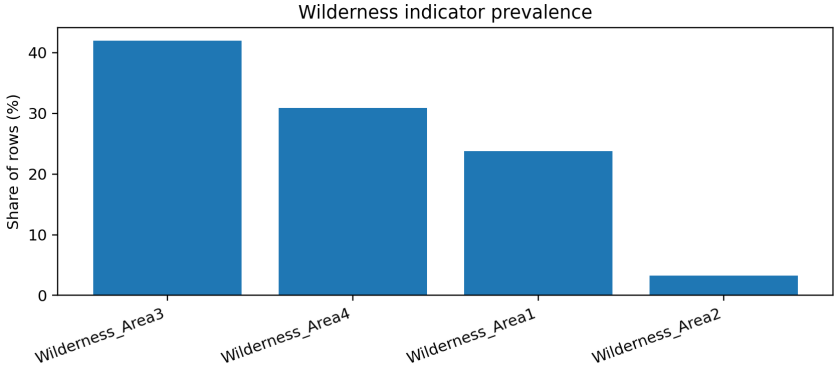


Figure 1: Marginal prevalence of wilderness indicators.

Figure 1 shows that wilderness areas are not equally represented in the labelled training set. Since wilderness implicitly captures geographic location, climate regime, and altitude patterns, uneven prevalence can influence how much signal each indicator contributes.

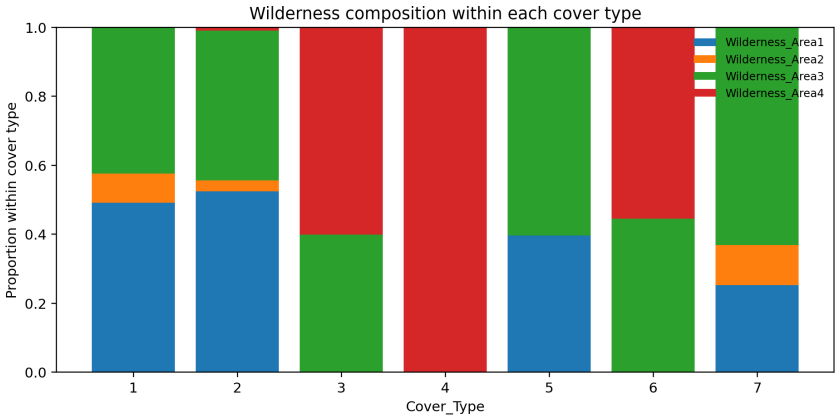


Figure 2: Wilderness composition within each Cover_Type (stacked proportions).

Figure 2 examines wilderness composition within each cover type. Several classes are concentrated in specific wilderness areas, indicating that these indicators function as compact location descriptors.

This supports the use of tree models, which can exploit high-order interactions between wilderness membership, elevation bands, and proximity features.

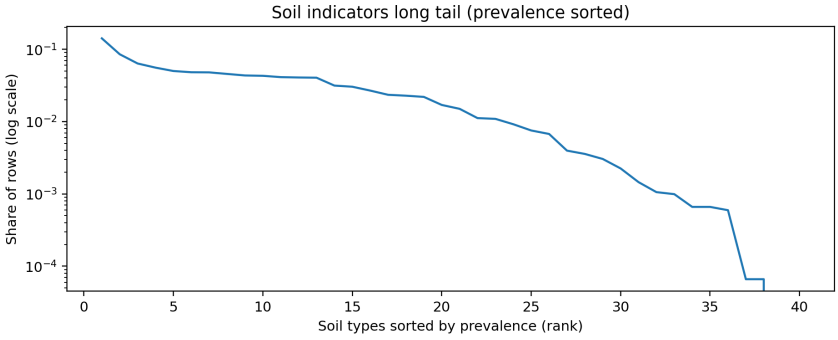


Figure 3: Soil indicator prevalence sorted by rank (log scale).

Figure 3 shows a strong long tail in soil indicators. A small number of soil types are common, while many are extremely rare. For modelling, this suggests that naively treating each soil one-hot column as an independent predictor can lead to fragile split rules for rare types. It also motivates decoding soil indicators into more compact variables (for example, climatic and geologic zones) and including interactions with elevation and hydrology.

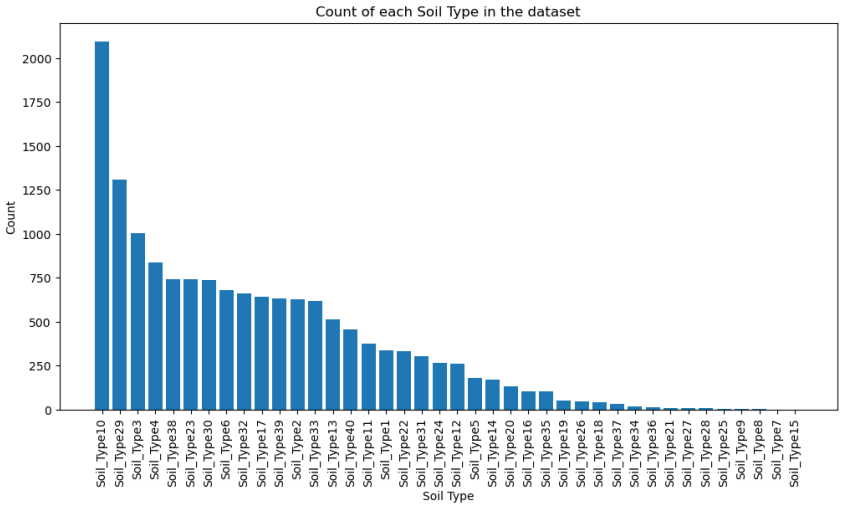
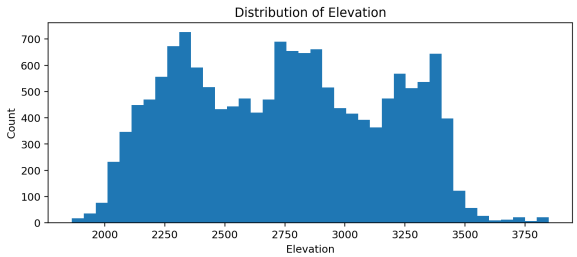


Figure 4: Topsoil indicators by frequency.

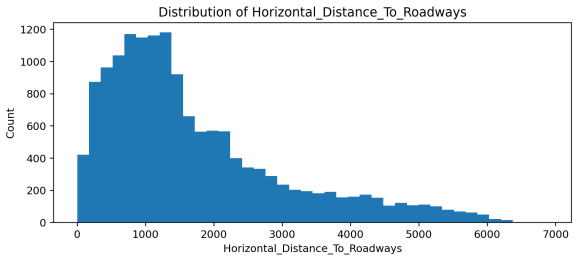
Figure 4 highlights the dominant soil indicators. These frequent soil types can carry substantial signal on their own and, combined with wilderness membership, often provide a strong prior over cover type.

3.5 Numeric Feature Ranges, Scale Differences, and Zeros

Continuous variables span very different numeric ranges. Elevation and distance to roads or fire points can reach several thousands, while slope and hillshade indices have much smaller ranges. This scale heterogeneity is important for linear and distance-based models, but less problematic for tree ensembles.



(a) Distribution of Elevation.



(b) Distribution of Horizontal_Distance_To_Roadways.

Figure 5: Distributions of key continuous features.

The EDA also reveals that hydrology distance variables contain a large mass at zero. We treat these zeros as meaningful geography rather than missingness. They indicate that a cell lies on a hydrology feature or directly adjacent to it. This motivates both keeping zeros unchanged and engineering a Euclidean distance to hydrology feature that combines horizontal and vertical components.

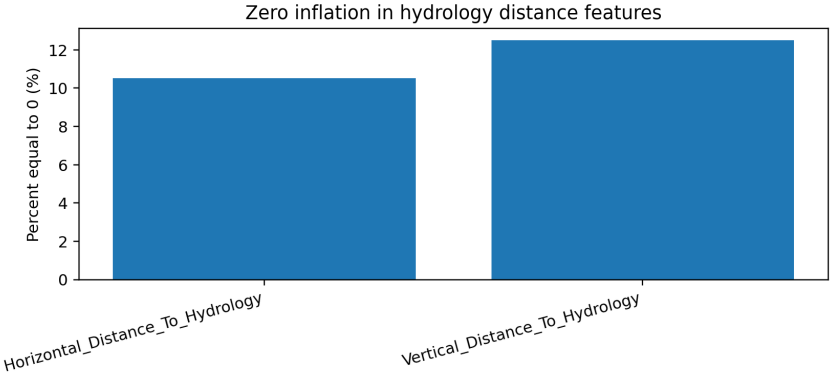
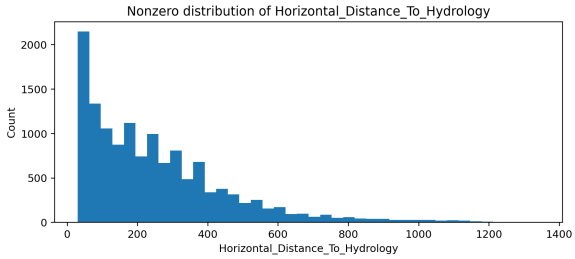
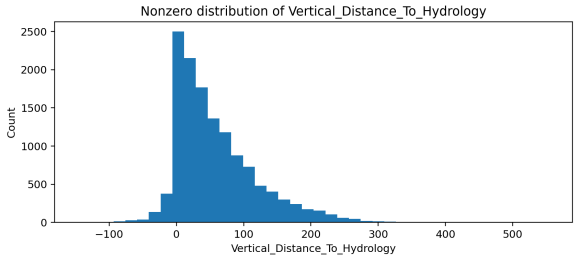


Figure 6: Zero inflation in hydrology distance features.



(a) Nonzero distribution of Horizontal_Distance_To_Hydrology.



(b) Nonzero distribution of Vertical_Distance_To_Hydrology.

Figure 7: Nonzero distributions of hydrology distance features.

Finally, Aspect is an angular variable. Its circular nature is not well captured by treating it as an ordinary numeric feature, while a polar histogram makes the cyclic structure explicit.

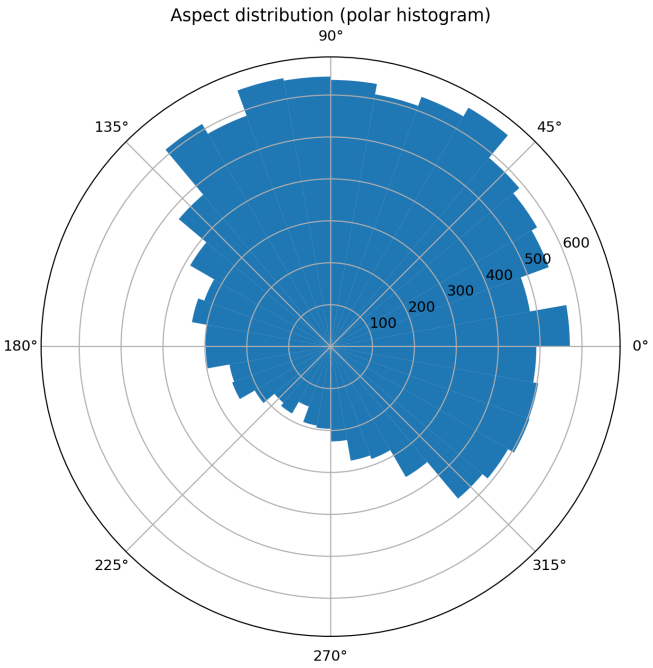
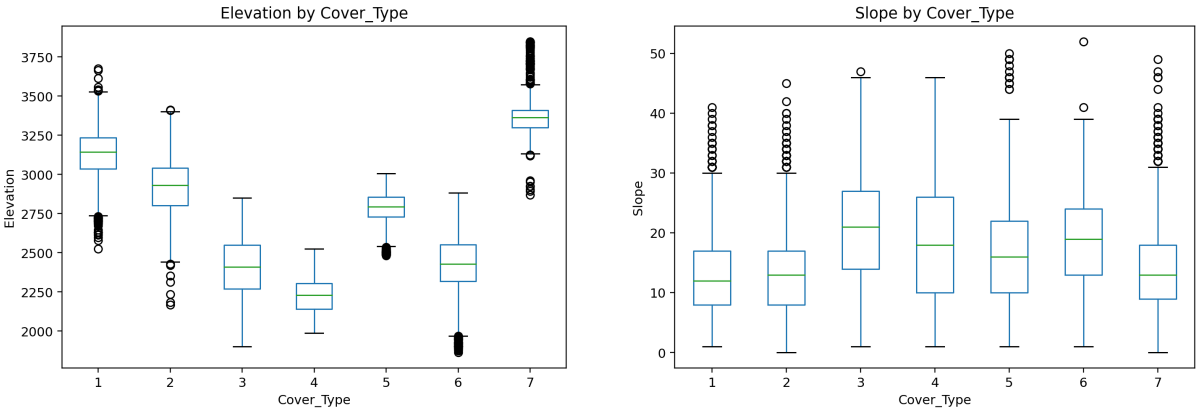


Figure 8: Aspect distribution (polar histogram).

To understand class separability, boxplots by cover type provide a compact summary. Elevation and slope show strong class-conditional structure, which suggests that non-linear decision boundaries and interaction terms will be useful.



(a) Elevation by Cover_Type.

(b) Slope by Cover_Type.

Figure 9: Boxplots of key features by cover type.

3.6 Correlation Structure and Modelling Implications

The profiling report highlights several strong correlations within the feature space. In particular, the three hillshade measurements (9 am, noon, and 3 pm) are highly correlated, reflecting that they capture related aspects of daily illumination under similar terrain conditions. Aspect also shows meaningful correlation with hillshade variables, which is expected since slope orientation directly influences sunlight exposure at different times of day. Additionally, horizontal and vertical distances to hydrology exhibit substantial correlation, as both measure proximity to the same physical feature along different axes.

These patterns suggest that variables describe overlapping geometric relationships, indicating that they encode shared terrain and exposure mechanisms rather than independent sources of information.

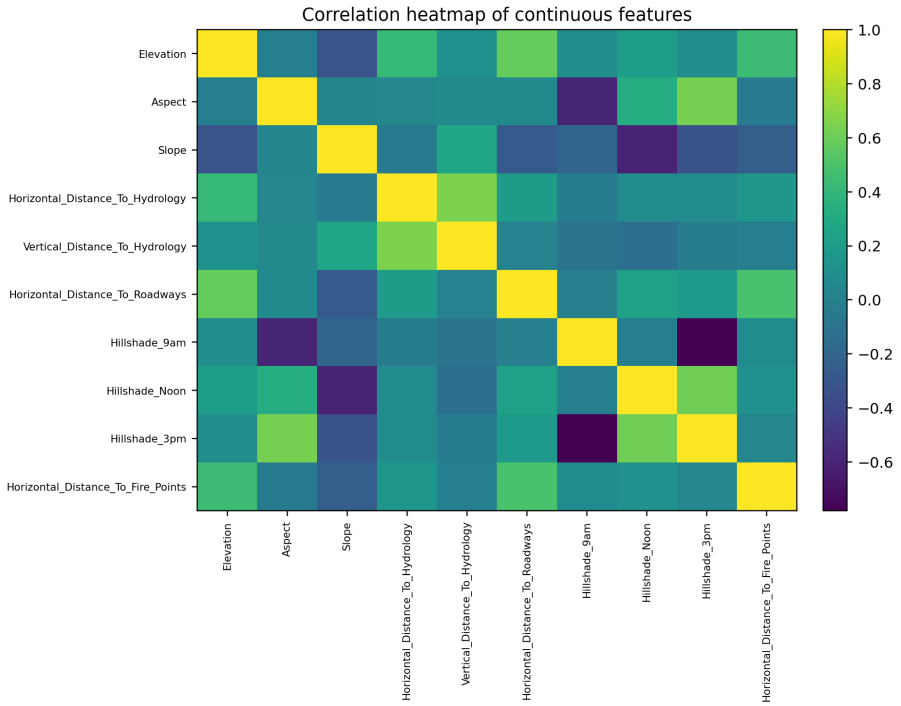


Figure 10: Correlation heatmap of continuous features.

4 Data Cleaning and Preparation

Our cleaning strategy is intentionally minimal because the data are already high quality. The exploratory analysis did not reveal missing values, duplicate rows, or data corruption; therefore, the preparation process focuses on structural consistency and data representation rather than corrective cleaning.

Removal of non-informative identifiers.
The `Id` column was removed from both training and test sets to prevent the model from learning information unrelated to the prediction task.

Removal of constant predictors.
`Soil_Type15` was dropped because it is constant zero and therefore provides no discriminative information.

Preservation of all observations.
No rows were removed, as the profiling report confirmed the absence of missing values and duplicates. Maintaining the full dataset preserves the empirical distribution and avoids introducing sampling bias.

Retention of extreme and zero values.
Outliers were not removed. In this context, extreme elevations, slopes, or distances correspond to legitimate terrain configurations rather than measurement noise. Since tree-based ensemble methods are robust to extreme values and linear models are regularised, removing these observations would risk discarding rare but potentially informative geographic regimes.

Deterministic feature engineering applied consistently.
All engineered features (e.g., Euclidean distance to hydrology, aggregated hillshade measures, circular encoding of aspect) are deterministic transformations computed row-wise. The same feature engineering function was applied to both train and test datasets to ensure column alignment and prevent data leakage.

Model-aware preprocessing strategy.
Statistical transformations that depend on data distribution (e.g., scaling for linear models) were performed within cross-validation pipelines to avoid leakage. Tree-based models were trained on raw features, as they are invariant to monotonic scaling and inherently capture nonlinear interactions.

Overall, the preparation phase prioritises preserving the original cartographic structure of the data while ensuring statistical validity and consistency with the modelling framework.

5 Feature Engineering

Feature engineering in this project aims to refine the representation of terrain characteristics, to better reflect ecological processes such as water proximity, sunlight exposure, and elevation-driven regimes.

All engineered variables are generated through a single function and applied identically to both training and test sets. Since these transformations are deterministic and operate row-wise, they maintain column alignment and avoid leakage across validation folds.

After removing non-informative columns and applying deterministic transformations, the final representation contains **71 predictors** (excluding the target).

5.1 Engineered Feature Inventory

Table 2: Engineered feature inventory.

Feature / Group	What It Captures	Why It Is Useful
Hillshade_mean	Average illumination across morning, noon, afternoon	Proxy for overall light exposure and shading conditions
Hillshade_range	Intra-day variation in hillshade	Captures terrain-orientation-driven light variability
Hillshade_Asymmetry	9am − 3pm hillshade	Encodes east vs. west exposure asymmetry explicitly
Hydrology_Euclidean	$\sqrt{h^2 + v^2}$ to hydrology	More physically meaningful water proximity than separate raw distances
Hydrology_Manhattan	$\|h\| + \|v\|$ hydrology distance	Alternative geometry that can help trees split on grid-like distance patterns
Water_Elevation	Estimated elevation of nearest water body	Combines elevation and vertical hydrology signal into a terrain reference level
Elevation_sq	Quadratic elevation term	Lets models capture non-linear elevation effects
Elevation_x_Slope	Interaction between elevation and slope	Represents combined terrain severity and exposure
Elev_Hydro_ratio	Elevation relative to hydrology proximity	Normalises elevation by water distance to reflect ecological gradients
Elevation_band	Binned ecological elevation zone (0–4)	Groups elevation into broad ecological zones
Slope_sq	Quadratic slope term	Captures non-linear slope effects and steeper threshold behaviour
Fire_Road_Sum	Total proximity signal for human access & ignition	Combines two relevant distance channels
Fire_Road_Diff	Absolute difference between fire and road distances	Separates locations near one source but far from the other
Aspect_sin, Aspect_cos	Circular encoding of aspect	Removes 0° vs. 360° discontinuity
Wilderness_Area_int	Integer-decoded wilderness zone	Compact representation of one-hot wilderness indicators
Wild_x_ElevBand	Wilderness zone × elevation band	Encodes region-specific altitude regimes
Soil_climatic_zone, Soil_geologic_zone	Decoded higher-level soil descriptors from ELU codes	Transforms sparse soil one-hot variables into interpretable ecological abstractions

5.2 Circular Encoding of Aspect

Aspect is circular; therefore, treating it as a linear variable introduces artificial discontinuity. For instance, the directions 359° and 1° are almost identical but look far apart with a linear representation. We encode Aspect using sine and cosine transformations and drop the raw

Aspect column:

$$\texttt{Aspect_sin} = \sin\big(\texttt{Aspect} \cdot \pi/180\big), \tag{1}$$
$$\texttt{Aspect_cos} = \cos\big(\texttt{Aspect} \cdot \pi/180\big). \tag{2}$$

This two-dimensional encoding fixes this discontinuity, preserving geometric structure and reducing representational bias.

5.3 Hydrology Feature Family

The raw dataset provides horizontal and vertical distances to hydrology separately. While informative, these represent orthogonal components of a single physical quantity: proximity to water. We construct:

Hydrology_Euclidean.

$$\texttt{Hydrology_Euclidean} = \sqrt{h^2 + v^2} \tag{3}$$

This captures true geometric distance and is the most interpretable proximity measure for many terrain processes.

Hydrology_Manhattan.

$$\texttt{Hydrology_Manhattan} = |h| + |v| \tag{4}$$

This alternative metric provides a monotonic decomposition that tree-based models can exploit efficiently through threshold splits.

Water_Elevation.

$$\texttt{Water_Elevation} = \texttt{Elevation} - \texttt{Vertical_Distance_To_Hydrology} \tag{5}$$

Water_Elevation estimates the elevation of the nearest water body, derived from elevation and vertical hydrology distance. This creates a reference level that can help distinguish forest types associated with ridges, valleys, and drainage patterns.

5.4 Illumination Aggregates

The three hillshade variables (morning, noon, afternoon) are strongly correlated. Rather than removing them and risking losing directional information, we construct:

- `Hillshade_mean`: overall light exposure.
- `Hillshade_range`: intra-day variation.
- `Hillshade_Asymmetry`: 9 am − 3 pm difference (east–west exposure).

This aggregation strategy: reduces redundancy, preserves physical meaning, controls variance in linear models, and provides interpretable threshold candidates for trees.

5.5 Terrain Severity and Nonlinear Effects

Some ecological processes vary nonlinearly with elevation and slope; hence we introduce selected, interpretable nonlinear terms:

- `Elevation_sq` ($\texttt{Elevation}^2$)

- `Slope_sq` ($\texttt{Slope}^2$)
- `Elevation_x_Slope` ($\texttt{Elevation} \times \texttt{Slope}$)

These additions allow linear models to approximate curved ecological gradients while maintaining controlled complexity. Importantly, we avoid general polynomial expansion, which would increase hypothesis-class dimensionality and risk overfitting.

Tree ensembles inherently model higher-order interactions; therefore, nonlinear terms are limited and domain-driven rather than combinatorial.

5.6 Soil and Wilderness Decoding

The dataset contains numerous one-hot encoded soil and wilderness indicators. While sparse, these variables carry ecological signal. Instead of removing rare soil types, we decode higher-level ecological descriptors.

For wilderness, we convert the one-hot block into an integer zone index and then interact it with elevation band. For soils, we decode the active soil type into ELU-based climatic and geologic zones.

This provides two advantages. First, it isolates higher-level ecological information that can generalise better than extremely sparse individual soil flags. Second, it preserves interpretability: a split on `Soil_climatic_zone` or `Wild_x_ElevBand` is easier to explain than a split on a rare one-hot soil variable.

5.7 Ablation Study

To quantify the contribution of the engineered features, we conducted a structured ablation study. The objective was to isolate the incremental impact of each feature group on predictive performance.

Experimental setup. All ablation experiments were evaluated using classification accuracy. To ensure reliable comparison across configurations, performance was estimated using stratified cross-validation on the training portion of the data. This provides a stable estimate of generalisation while preserving the original class proportions in each fold.

To isolate the impact of feature engineering, we fixed the learning algorithm across all experiments, using the baseline LightGBM configuration, identified during preliminary analysis as one of the most competitive models. Hyperparameters were kept constant for every ablation step, ensuring that performance differences reflect changes in the feature set rather than variations in model complexity or optimisation.

The baseline configuration includes only the cleaned original predictors (after removing `Id` and `Soil_Type15`), with no engineered features. Feature groups were then added incrementally:

1. Geometric encoding (`Aspect_sin`, `Aspect_cos`)
2. Hydrology family (Euclidean, Manhattan, Water_Elevation)
3. Hillshade aggregates (mean, range, asymmetry)
4. Terrain nonlinear terms (`Elevation_sq`, `Slope_sq`, `Elevation_x_Slope`)
5. Ecological grouping (`Elevation_band`, Wilderness decoding, soil climatic/geologic zones)

Table 3: Ablation study results.

Configuration	Features	CV Accuracy (mean ± std)	Δ Baseline
Baseline (cleaned only)	53	0.8655 ± 0.0040	—
+ Aspect encoding	54	0.8626 ± 0.0080	-0.0029
+ Hydrology family	57	0.8697 ± 0.0084	$+0.0042$
+ Hillshade aggregates	60	0.8686 ± 0.0064	$+0.0031$
+ Terrain nonlinear terms	66	0.8786 ± 0.0062	$+0.0131$
+ Ecological grouping (full)	71	0.8769 ± 0.0085	$+0.0114$

Results. The ablation study reveals that not all engineered feature groups contribute equally. The most substantial improvement arises from terrain-based nonlinear transformations (`Elevation`^{2} and `Elevation` × `Slope`), which increase cross-validated accuracy by more than 1.3 percentage points relative to the baseline. Hydrology-derived features provide a moderate but consistent improvement, confirming the importance of geometric water proximity. In contrast, circular aspect encoding does not improve performance for tree-based models, suggesting that gradient boosting can already capture directional splits effectively from the raw variable. Ecological grouping features provide minimal incremental benefit once terrain nonlinearities are included, indicating that much of the ecological structure is already captured by elevation-driven transformations.

We retained ecological grouping features despite the small accuracy drop observed in the ablation. The difference (-0.17%) falls within the standard error of the CV estimate, so there is no strong statistical basis for exclusion. These features also serve a secondary purpose: they encode interpretable ecological abstractions such as climatic zone and wilderness–elevation interactions, which aid model explanation. Finally, the ablation used a fixed baseline LightGBM configuration, and the contribution of individual feature groups may differ once hyperparameters are tuned or models are combined in an ensemble.

Based on the ablation results, we proceed with a feature set that includes all engineered feature groups except circular aspect encoding. This configuration achieves a cross-validated accuracy of 0.8791 ± 0.0055, which represents a substantial improvement over the cleaned baseline (0.8655) while maintaining stable variance across folds.

6 Validation Strategy and Optimisation Design

The validation strategy relies on a two-level setup designed to separate model tuning from final labelled holdout evaluation while keeping runtime practical.

First, the labelled dataset is split once into an 80% training portion and a 20% holdout set, stratified by `Cover_Type`. The holdout set is never used during tuning and is reserved for the final comparison of the selected models.

Hyperparameter optimisation is performed only on the 80% training portion. For computational reasons, Optuna evaluates each trial on an internal train/validation split of this training set, again stratified by class. In other words, Optuna optimises validation accuracy on a single validation split rather than running cross-validation inside each trial. This choice substantially reduces runtime when tuning multiple strong models, but it also makes the objective noisier than CV-based tuning. This asymmetry is intentional: feature selection is a one-time decision affecting all downstream models, so we invested in CV stability there, whereas hyperparameter tuning involves hundreds of trials where noise is partially averaged out across the search trajectory. To reduce sensitivity to a single split, we fix the random seed and keep the split stratified; in addition, final model selection is confirmed using the outer holdout evaluation.

After tuning, individual models and the soft voting ensemble are evaluated on the outer holdout with accuracy, classification report, and confusion matrix.

Finally, the selected model is trained on the full labelled dataset to produce the competition submission.

7 Model Selection

This section describes how the final ensemble was selected. The process followed four steps: baseline comparison, candidate shortlisting, hyperparameter optimisation, and ensemble construction. Keeping this chronology explicit is important, because several different evaluation procedures are used for different purposes.

First, all candidate models were compared on the same labelled split: 80% of the data were reserved for model development and 20% were held out for final labelled evaluation. Within the 80% development split, baseline comparison was performed using stratified cross-validation. This stage was used only to rank model families and identify promising candidates.

Second, once the strongest model families had been identified, hyperparameter optimisation was performed only on the 80% development split. For computational reasons, Optuna evaluated each trial on a single stratified inner train–validation split rather than running full cross-validation inside every trial. This makes the tuning stage faster, while the outer holdout set remains untouched and available for final comparison.

Third, after tuning, the selected models were retrained on the full development split and compared on the 20% holdout set. The holdout results were used to decide which models entered the final ensemble.

Finally, the chosen models were combined through soft voting, which averages class-probability vectors rather than hard class labels. This is appropriate here because the strongest candidates are all probabilistic tree ensembles, but they do not make exactly the same errors.

7.1 Baseline Evaluation: Model Comparison

We began with a broad benchmark across linear models and tree ensembles, all evaluated under the same pipeline and data split. The purpose of this stage was not to optimise each model exhaustively, but to establish which model families were competitive enough to justify further work.

Table 4: Baseline model comparison.

Model	**CV Accuracy (mean ± std)**
Dummy (majority)	0.1429
Logistic Regression (L2)	0.7015 ± 0.0046
Logistic Regression (Elastic Net)	0.7021 ± 0.0046
Random Forest	0.8285 ± 0.0069
ExtraTrees	0.8248 ± 0.0062
HistGradientBoosting	0.8308 ± 0.0018
XGBoost	0.8405 ± 0.0023
LightGBM	0.8501 ± 0.0020

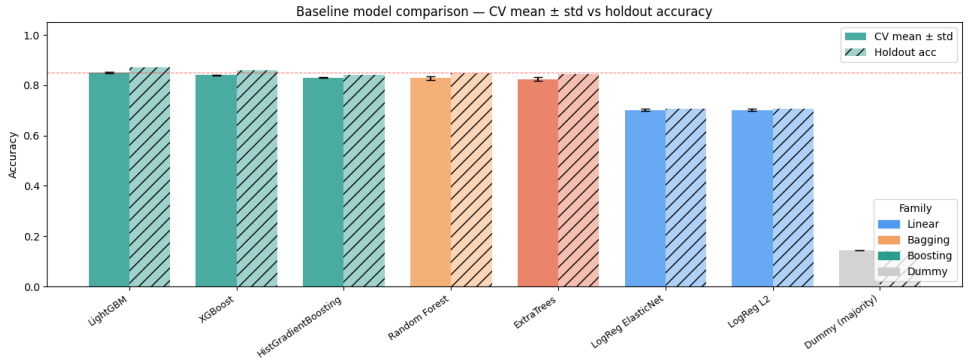


Figure 11: Baseline model comparison (CV and holdout accuracy).

The comparison produced a clear pattern. Logistic regression remained far behind the tree-based methods, with accuracy around 70%, which indicates that the class boundaries in this dataset are strongly nonlinear. This is consistent with the ecological structure of the problem: cover type depends on interactions among elevation, slope, hydrology, and sparse soil indicators that cannot be captured well by a linear decision boundary.

The tree ensembles performed substantially better. Among the initial candidates, LightGBM delivered the strongest overall baseline performance, followed by XGBoost and HistGradientBoosting. ExtraTrees and Random Forest were slightly weaker as single models, but they remained interesting because bagging-based ensembles often make different mistakes from boosting-based ensembles. Since diversity of errors is valuable in a soft-voting system, we did not evaluate models only by standalone accuracy.

This baseline comparison therefore served two purposes. It identified LightGBM and XGBoost as the strongest individual learners, and it suggested that ExtraTrees could still be useful as a complementary ensemble member even if its standalone score was lower.

7.2 Candidate Shortlisting and Initial Screening

After the baseline comparison, we reduced the set of models carried forward. HistGradientBoosting was competitive, but it consistently trailed LightGBM and XGBoost on both cross-validation and holdout accuracy. Because its predictive profile was also closer to the other boosting models than to the bagging models, it offered less additional diversity for ensembling.

For that reason, the final shortlist contained three models: LightGBM, XGBoost, and ExtraTrees. The first two were retained because they were the strongest single-model performers. ExtraTrees was retained for a different reason: although slightly weaker on its own, it introduces a different bias–variance tradeoff and a different source of randomness, which makes it a plausible complement to the boosted trees in a probability-averaging ensemble.

This shortlisting step is important because it separates two questions that are often conflated. The best individual model is not necessarily the best ensemble component, and the best ensemble is not necessarily composed only of the top standalone models.

7.3 Final Ensemble Candidates: LightGBM, XGBoost, and ExtraTrees

The final ensemble candidates therefore came from two distinct families. LightGBM and XGBoost are boosting models that build trees sequentially, so each new tree focuses on correcting the residual errors of the current ensemble. This typically produces strong accuracy when the signal is nonlinear and tabular, as in this dataset.

ExtraTrees, by contrast, belongs to the bagging family. Instead of building a sequence of corrective trees, it reduces variance by averaging over many highly randomised trees. In addition to bootstrap-style randomness and feature subsampling, ExtraTrees randomises the split thresholds themselves. This generally makes its individual predictions noisier than those of the boosting models, but it can also make its errors less correlated with theirs.

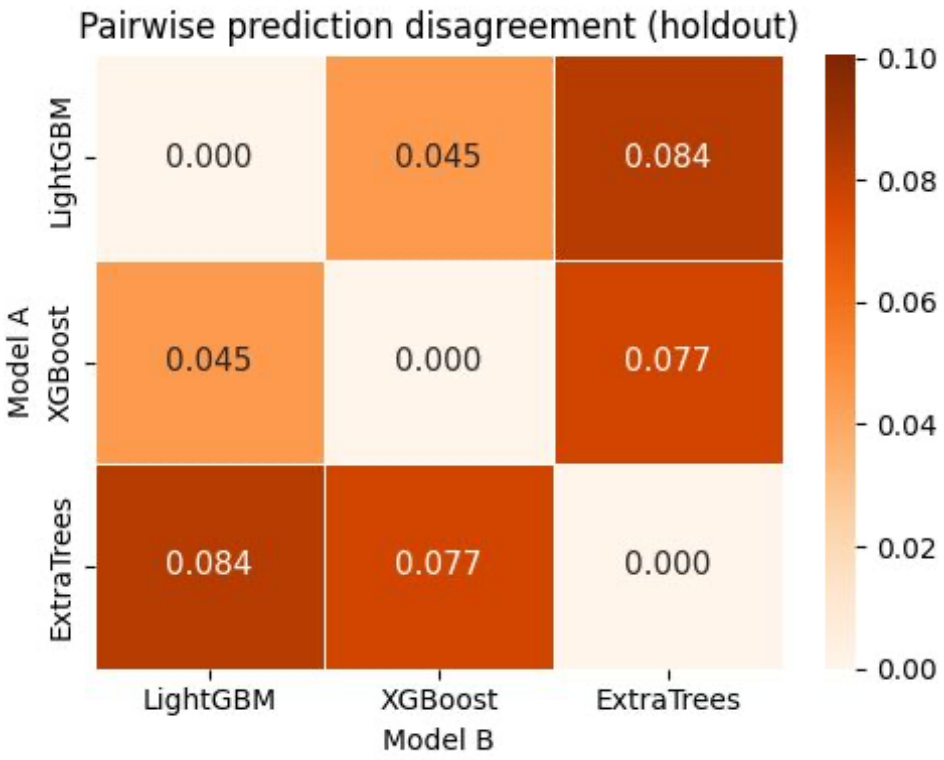


Figure 12: Error complementarity among the three ensemble candidates.

That difference matters for ensembling. If two models are both strong but tend to fail on the same examples, averaging them brings limited benefit. If a third model is slightly weaker but fails on different examples, its contribution to soft voting may still be positive. For this reason, model complementarity was treated as a selection criterion alongside standalone accuracy.

7.4 TabICL: Transformer-Based In-Context Learning (Optional)

In addition to the three tree-based candidates, we also experimented with TabICL, a transformer-based in-context learning approach for tabular classification. This model is conceptually different from the rest of the pipeline and therefore serves a different purpose in the study.

We were particularly drawn to the recent work by Le Morvan et al. (Inria), as foundation models for tabular data represent an emerging field with significant potential. This interest is further supported by Zabërgja et al. (2025), who conduct a large-scale benchmark comparing deep learning, classical ML, AutoML, and foundation model approaches across a broad range of tabular classification tasks.

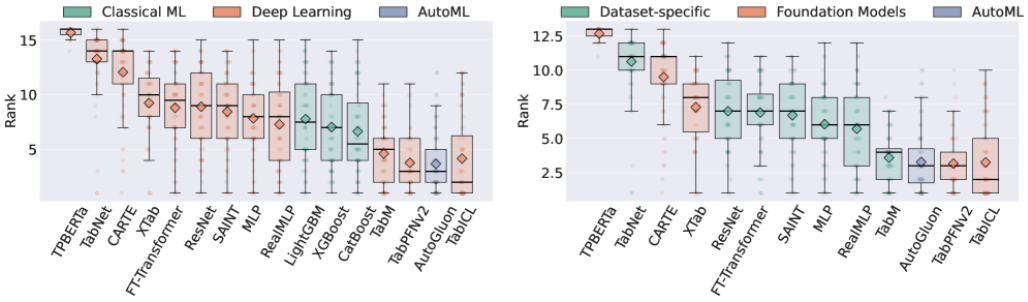


Figure 13: **Left:** Distribution of ranks for Deep Learning, Classical ML, and AutoML methods by classifier family. **Right:** Distribution of ranks for Foundation Models, Dataset-Specific methods, and AutoML methods by classifier family.

TabICL is a transformer-based foundation model for tabular data that operates through in-context learning: given a set of labelled training examples as context, it predicts labels for new queries directly, without any gradient-based fine-tuning. This represents a fundamentally different paradigm from gradient boosting or bagging. Pre-trained on millions of synthetic tabular datasets, the model learns to recognise statistical patterns between features and labels from context examples alone, analogous to how large language models perform new tasks from few-shot prompts.

TabICL was not included as a core component of the final labelled-validation pipeline for two reasons. First, its behaviour is less transparent than that of the tuned tree ensembles, which makes it harder to interpret and calibrate within the rest of the workflow. Second, its probability outputs are not directly comparable to those of the tree models without additional care, since deep architectures often produce sharper and more overconfident predictions.

Nevertheless, TabICL produced a competitive leaderboard score of approximately 0.85 when submitted independently, which is notable given that it required no gradient-based training on this dataset. This confirms that foundation models for tabular data are approaching the performance of well-tuned tree ensembles on structured classification tasks. However, because the three-model ensemble achieved a higher score (0.86) and offers greater transparency and calibration control, we treat it as the primary deliverable. TabICL is reported here as a standalone alternative that demonstrates the viability of in-context learning for this problem class.

7.5 Soft-Voting Ensemble and a Unified Probability Interface

The final labelled pipeline combines LightGBM, XGBoost, and ExtraTrees through soft voting. Each model outputs a vector of class probabilities, and the ensemble prediction is obtained by averaging these vectors component-wise and then selecting the class with the highest averaged probability.

This choice is preferable to hard voting in our setting. Hard voting discards information about prediction confidence and only retains the top predicted class from each model. Soft voting preserves the full probability distribution, which is more appropriate when the strongest models are already probabilistic classifiers and when some class distinctions are intrinsically ambiguous.

A practical requirement is that all ensemble members must expose probabilities in the same class order, namely classes 1 through 7. This was enforced explicitly in the prediction pipeline before averaging.

The rationale for soft voting here is not that averaging automatically guarantees better calibration. Rather, the expected benefit is variance reduction through error diversification. The boosted models provide the strongest single-model performance, while ExtraTrees contributes a

different error profile. The ensemble therefore aims to improve robustness by combining accurate but partially complementary predictors.

7.6 Hyperparameter Optimisation with Optuna (TPE)

Once the final three candidates had been selected, we tuned them with Optuna using the Tree-structured Parzen Estimator sampler. Optuna was chosen instead of `RandomizedSearchCV` because it uses the history of completed trials to guide future proposals toward more promising regions of the search space. This is particularly useful when the search space is mixed and moderately large, as in our three tree-based models.

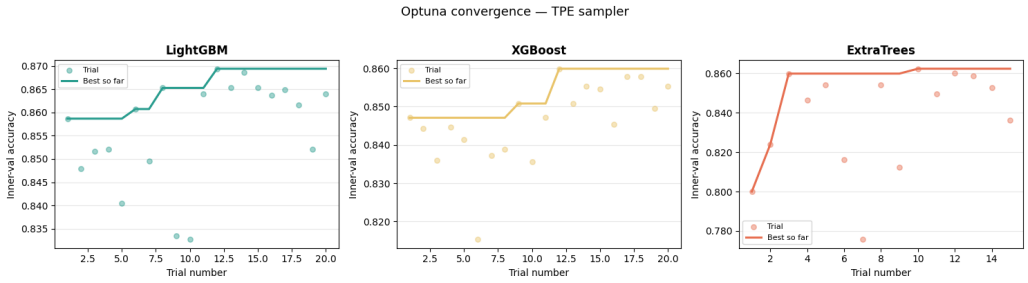


Figure 14: Optuna hyperparameter search overview.

The tuning budget was set to 20 trials for LightGBM, 20 trials for XGBoost, and 20 trials for ExtraTrees. Each trial was evaluated on a stratified inner train–validation split built from the 80% development portion of the labelled data. In other words, cross-validation was used for baseline model comparison, whereas Optuna tuning relied on a single inner validation split for speed. The 20% outer holdout set remained untouched throughout this procedure.

The search spaces were chosen to cover the parameters that most strongly control model capacity, regularisation, and stochasticity. For LightGBM and XGBoost, this includes the number of trees, learning rate, tree depth, subsampling, column subsampling, and regularisation parameters. For ExtraTrees, the main levers are the number of trees, maximum depth, feature subsampling, and minimum leaf size.

After optimisation, the best configuration of each model was retrained on the full development split and evaluated on the holdout set. These tuned versions were then used as the components of the final soft-voting ensemble.

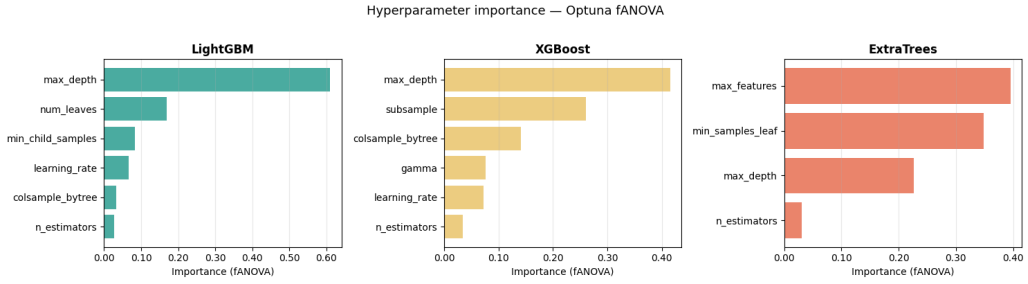


Figure 15: Optuna convergence plots for the three ensemble members.

8 Final Model, Performance, and Conclusion

8.1 Final Model

The final labelled model is a soft-voting ensemble composed of three tuned classifiers: LightGBM, XGBoost, and ExtraTrees. All three models were trained on the same engineered feature set selected through the ablation study, namely the cleaned original predictors plus hydrology features, hillshade aggregates, terrain nonlinearities, and ecological grouping variables, while excluding circular aspect encoding.

This final architecture reflects two complementary design principles. First, the ensemble includes the strongest individual learners identified during model comparison, namely the two gradient-boosting models. Second, it also includes a more randomised bagging model, ExtraTrees, because diversity of errors is valuable when averaging class probabilities.

8.2 Final Performance and Error Analysis

On the labelled evaluation pipeline, the final ensemble achieved strong overall accuracy and transferred well to the Kaggle test environment, where it obtained a public leaderboard score of 0.86. The gap between internal labelled performance and public leaderboard performance remains moderate, which suggests that the model generalises reasonably well and does not rely on overly fragile patterns.

Class-level performance is not uniform, but it is consistently strong. The clearest classes are those with distinctive elevation and wilderness signatures, especially Classes 4 and 7, which achieve very high precision and recall. The main remaining difficulty lies in separating Classes 1 and 2, whose ecological profiles overlap more strongly in elevation range and wilderness composition. This is a known challenge of the dataset and remains visible even in the final ensemble.

The confusion matrix therefore supports the same conclusion as the aggregate metrics: the model performs well across all seven classes, but the residual error is concentrated in ecologically similar cover types rather than being spread arbitrarily.

8.3 Conclusion

This project developed a complete machine learning pipeline for the Forest Cover Type classification task, from exploratory data analysis to feature engineering, model comparison, hyperparameter optimisation, and final ensembling.

Three findings are especially important. First, tree-based models are decisively better suited than linear models for this problem, which confirms that the class boundaries are strongly nonlinear. Second, feature engineering still matters even in tabular tree ensembles: the ablation study shows that terrain nonlinearities and hydrology-based variables provide meaningful improvements over the cleaned baseline. Third, the best final solution is not simply the best single model, but a combination of strong and complementary models whose errors are not perfectly aligned.

Relative to the earlier version of the project, the performance gain is real but not the main achievement. The more important improvement is methodological: the final pipeline is now supported by explicit validation logic, a documented ablation study, and a clear rationale for each component retained in the model. In that sense, the project progressed not only toward a better score, but toward a more defensible and interpretable modelling process.